Akshat Gurbuxani

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Two years of industry experience as an ML Engineer. Working with LLMs. Seeking Data Science and ML Engineer roles.

Experience

AI Engineer April 2025 – Present
Semiotic Labs *Los Altos, LA*
- Trained **SOTA** multi-modal **NFT spam detection** system across **8** blockchain networks using **fine-tuned LLM**, **GNN**, **XGBoost** with **Mixture-of-Experts** ensemble, achieving **87%** accuracy, correctly flagging **150,000+** malicious contracts
- Built trajectory-aware **multi-agent** pipeline with **DSPy ReAct agents**, integrating The Graph's Token API and Subgraphs with judge-agent for evaluation; improved response accuracy by **70%**, reduced query latency by **60%** in production

Data Science Intern Sep. 2024 – Dec. 2024
Boston Public School *Boston, MA*
- Designed **RAG**-based chatbot with BPS policy repository, automating document checks and reducing manual effort by **85%**
- Aligned **FAISS** with vector and lexical search, metadata-driven clustering, and a **multi-agent LLM** pipeline, enhancing retrieval speed by **70%** and document relevance by **90%**

Machine Learning Engineer July 2021 – Aug. 2023
Optum Global Solutions Pvt Ltd *Hyderabad, India*
- Productionized neural collaborative filtering model to personalize insurance policy recommendations, increasing retention by **70%** and page traffic by **86%**. Integrated **Kafka** and **RESTful APIs** for real-time content delivery
- Optimized ML pipelines using **Apache Spark**, reducing model training time by **75%**, enabling faster model iterations and scalable deployments
- Developed **LSTM**-based time-series forecasting model for website traffic prediction, attaining **80%** accuracy. Automated testing workflows with **pytest**, **Docker**, **CI/CD**, **Jenkins**, and **Kubernetes** reducing manual testing by **70%**
- Engineered distributed data pipeline using **Apache Kafka** and **Apache Flink** for real-time data processing, reducing data latency from **40 minutes to 5 minutes**, and increasing data throughput by **80%**
- Structured multi-node model deployment architecture using **Kubernetes** and **Docker**, enabling **parallel inference** across multiple **GPUs**, reducing inference time by **70%**, supporting high-throughput applications

Education

Boston University Boston, MA
Master of Science in Artificial Intelligence *Sep. 2023 – Dec. 2024*

Graduate Teaching Assistant: *Grad Intro to Natural Language Processing (NLP), Data Science Tools and Applications*
Related Coursework: *Computer Vision, Deep Learning, Reinforcement Learning, Machine Learning, Medical Imaging with DL*

SRM Institute of Science and Technology Chennai, India
Bachelor of Science in Computer Science *Aug. 2017 – May 2021*

Publications

Solving International Mathematical Olympiad at Human Level June 2024 – Aug. 2024
- Co-authored paper on optimizing **Large Language Models** for olympiad- and PhD-level math reasoning
Under review at ICLR 2025

Technical Skills

Programming & Data Science: Python (NumPy, Pandas, Matplotlib, scikit-learn), C#, C++, FAISS, SQL, MongoDB
Machine Learning: LLM (RNN, CNN, RLHF, Transformer), TensorFlow, PyTorch, spaCy, LangChain, MLFlow, CUDA
Big Data & MLOps: Hadoop, Apache (Spark, Kafka, Flink), MLOps, AWS (SageMaker, EC2)
Software Engineering & Deployment: Docker, Kubernetes, CI/CD, Jenkins, FastAPI, Streamlit, Git

Projects

AI Mathematical Olympiad | *Python, LLM, Prompt Engineering* | April 2024 – June 2024
- Secured **silver** medal in Kaggle's AIMO by applying **Chain-of-Thought** reasoning with self-consistency, fine-tuning, and optimizing **DeepSeekMath LLM** to solve olympiad-level math problems

AI Stock Trader | *Python, Transformer, LLM, Predictive Modeling* | Jan. 2024 – May 2024
- Engineered **LLM-based** trading pipeline leveraging real-time news and historical stock data to automate **E-mini S&P 500** futures trades, yielding **$1M+** in simulated revenue

Itinerary Planner | *Python, Transformer, LLM, Recommender Systems* | Sep. 2024 – Nov. 2024
- Built trip planner app with **Gemini** and real-time data pipelines, generating **personalized itineraries** based on dates, budget, and preferences